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| <b>DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE</b><br><b>Summer Examination – 2023</b><br><b>Course: T.Y.B. Tech.                      Branch : Mech. Engg.                      Semester :VI</b><br><b>Subject Code &amp; Name: BTMPE604D      Robotics</b><br><b>Max Marks: 60                      Date:      19/07/2023                      Duration: 3 Hr.</b>   |                                                                                                                                         |              |           |
| <b>Instructions to the Students:</b><br>1. All the questions are compulsory.<br>2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.<br>3. Use of non-programmable scientific calculators is allowed.<br>4. Assume suitable data wherever necessary and mention it clearly. |                                                                                                                                         |              |           |
|                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                         | (Level/CO)   | Marks     |
| <b>Q. 1</b>                                                                                                                                                                                                                                                                                                                                                                        | <b>Solve Any Two of the following.</b>                                                                                                  |              | <b>12</b> |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                          | What is Industrial Automation ? Discuss various types of Industrial automation                                                          | <b>U/CO1</b> | <b>6</b>  |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                          | Explain various robot configurations with sketches.                                                                                     | <b>U/CO1</b> | <b>6</b>  |
| <b>C)</b>                                                                                                                                                                                                                                                                                                                                                                          | Write short note on Robotics control system                                                                                             | <b>U/CO1</b> | <b>6</b>  |
|                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                         |              |           |
| <b>Q.2</b>                                                                                                                                                                                                                                                                                                                                                                         | <b>Solve Any Two of the following.</b>                                                                                                  |              | <b>12</b> |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                          | Derive the forward kinematic equation for 2 DOF robot manipulator having link length L1, L2 & Orientation angle $\theta_1$ , $\theta_2$ | <b>E/CO2</b> | <b>6</b>  |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                          | Write and explain Homogeneous transformation matrix for rotation and Translation.                                                       | <b>U/CO2</b> | <b>6</b>  |
| <b>C)</b>                                                                                                                                                                                                                                                                                                                                                                          | Write a short note on Trajectory planning in robotics.                                                                                  | <b>U/CO2</b> | <b>6</b>  |
|                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                         |              |           |
| <b>Q. 3</b>                                                                                                                                                                                                                                                                                                                                                                        | <b>Solve Any Two of the following.</b>                                                                                                  |              | <b>12</b> |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                          | Distinguish between Hydraulic, Pneumatic and Electrical Drives.                                                                         | <b>E/CO3</b> | <b>6</b>  |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                          | Describe construction and working of AC Motors                                                                                          | <b>U/CO3</b> | <b>6</b>  |
| <b>C)</b>                                                                                                                                                                                                                                                                                                                                                                          | Describe construction and working of Stepper Motors                                                                                     | <b>U/CO3</b> | <b>6</b>  |
|                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                         |              |           |
| <b>Q.4</b>                                                                                                                                                                                                                                                                                                                                                                         | <b>Solve Any Two of the following.</b>                                                                                                  |              | <b>12</b> |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                          | Describe construction and working of LVDT to measure the linear displacement.                                                           | <b>E/CO4</b> | <b>6</b>  |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                          | Describe construction and working of Proximity Sensors                                                                                  | <b>U/CO4</b> | <b>6</b>  |
| <b>C)</b>                                                                                                                                                                                                                                                                                                                                                                          | Explain Magnetic gripper in details                                                                                                     | <b>U/CO4</b> | <b>6</b>  |
|                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                         |              |           |
| <b>Q. 5</b>                                                                                                                                                                                                                                                                                                                                                                        | <b>Solve Any Two of the following.</b>                                                                                                  |              | <b>12</b> |

|           |                                                                             |              |          |
|-----------|-----------------------------------------------------------------------------|--------------|----------|
| <b>A)</b> | Explain advantages and disadvantages of lead through programming in details | <b>U/CO5</b> | <b>6</b> |
| <b>B)</b> | Discuss economic analysis of Robots                                         | <b>U/CO5</b> | <b>6</b> |
| <b>C)</b> | Explain the role of robot in material handling                              | <b>U/CO5</b> | <b>6</b> |
|           | <b>*** End ***</b>                                                          |              |          |

**The grid and the borders of the table will be hidden before final printing.**